

Divyam Choudhary

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EDUCATION

Vellore Institute of Technology, Bhopal

2023 - 2027

B.Tech - Computer Science and Engineering - CGPA: 8.44

Bhopal, Madhya Pradesh

Prudence School

2021 - 2023

Higher Secondary: 77% (2022-23) | Matriculation: 78% (2021-22)

New Delhi, Delhi

EXPERIENCE

Amazon ML Summer School 2025

Aug 2025 – Sep 2025

Mentee

Virtual

- Selected among a small number of students from VIT Bhopal to participate in **Amazon's ML Summer School**.
- Received hands-on training from Amazon scientists on **bias-variance tradeoff, regularization, and ensemble methods**, alongside supervised and unsupervised learning fundamentals.
- Implemented machine learning algorithms using **Python, Scikit-learn, NumPy, and Pandas** through practical assignments and case studies.

PROJECTS

Locust Watch - Outbreak Risk Predictor | **Python, Scikit-learn, FastAPI, React, Leaflet**

- Shipped a live, full-stack ML web app that lets you click any point on the map and get a real-time desert-locust swarm-risk score. **React** frontend (Netlify) + **FastAPI** backend (Render), live at locust-watch.netlify.app.
- Built the dataset from scratch, fusing **2,500+ FAO swarm records** with **TerraClimate climate data** extracted per-coordinate from NetCDF grids; no pre-made dataset.
- Caught my own model lying: random split scored **0.84 AUC**; **spatial cross-validation** exposed it as leakage and revealed the honest **0.76 AUC**, catching ~79% of real swarms.
- Served it as a real API. Validated **FastAPI /predict** endpoint returning per-feature attribution (why each score), consumed by an interactive map, sliders, and risk gauge.

Drug Discovery AI Pipeline | **Python, RDKit, Scikit-learn**

- Built and validated a molecular property prediction pipeline on the **MoleculeNet BBBP dataset** (2,050 compounds), generating Morgan fingerprints via **RDKit** and training a **Random Forest classifier** to predict blood-brain barrier permeability.
- Exposed a **0.05 AUC optimism gap** between naive random splitting (0.94 AUC) and scaffold-based splitting (0.88 AUC), replicating a known methodological rigor pattern from prior spatial cross-validation work.
- Diagnosed and fixed a **class-imbalance failure mode** in scaffold splitting: a size-sorted allocation isolated the minority class entirely into training, resolved via shuffled scaffold-group assignment.

TRAINING & ACHIEVEMENTS

- **Google Cloud Generative AI Certification** - Score: 92.6/100
- **NPTEL: Introduction to Machine Learning** (IIT Madras)
- Completed **The Bits and Bytes of Computer Networking** by Google.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL, HTML, CSS

Web: React.js, Vite, Leaflet, Node.js, Express.js, FastAPI, REST APIs, MongoDB

ML / Data: Scikit-learn, Pandas, NumPy, xarray; feature engineering, model evaluation, spatial cross-validation

Deployment / Tools: Render, Netlify, Git/GitHub, Postman, Google Earth Engine

Concepts: DSA, OOP, client-server architecture, MVC